

Python Data Types — 25 MCQs

Topics: int • float • complex • bool • str • list • tuple
Logic • Output Prediction • Mutability • Slicing • Identity • Type Behavior

Q1. What is the output?

```
x = [1, 2, [3, 4]]  
y = x[:]  
y[2].append(5)  
print(x, y, x is y)
```

- A. [1, 2, [3, 4]] [1, 2, [3, 4, 5]] False
- B. [1, 2, [3, 4, 5]] [1, 2, [3, 4, 5]] False
- C. [1, 2, [3, 4, 5]] [1, 2, [3, 4]] False
- D. [1, 2, [3, 4, 5]] [1, 2, [3, 4, 5]] True

Q2. Which statement is correct?

```
x = ([], )  
x[0].append(10)
```

- A. TypeError occurs because tuples are immutable.
- B. It is valid because the tuple's element reference is unchanged.
- C. The tuple automatically becomes a list.
- D. append creates a new tuple.

Q3. What is printed?

```
x = True + True * 2  
print(x, type(x))
```

- A. 3
- B. 3
- C. 2
- D. True

Q4. Which expression creates a tuple containing exactly one element?

- A. (42)
- B. (42,)
- C. tuple(42)
- D. [42]

Q5. What is printed?

```
x = (1, 2, 3, 4)  
print(x[-1], x[-3:-1])
```

- A. 4 (2, 3)
- B. 4 (3, 4)
- C. 3 (2, 3)
- D. 3 (3, 4)

Q6. Which statement is correct about strings?

- A. Strings can be modified character-by-character.
- B. Strings are mutable sequences.
- C. Strings are immutable sequences.
- D. A string can contain only ASCII characters.

Q7. What is printed?

```
s = 'PYTHON'  
print(s[1:5:2], s[::-1])
```

- A. YH NOHTYP
- B. YH NOHTYP
- C. YH NOHTYP
- D. YT NOHTYP

Q8. What is printed?

```
a = [1, 2]  
b = a  
c = a[:]  
b.append(3)  
print(a, b, c, a is b, a is c)
```

- A. [1,2,3] [1,2,3] [1,2] True False
- B. [1,2] [1,2,3] [1,2] False False
- C. [1,2,3] [1,2,3] [1,2,3] True True
- D. [1,2,3] [1,2] [1,2] False True

Q9. Which can be used as an element of a tuple?

- A. Only immutable values
- B. Only numbers
- C. Any valid Python object
- D. Only strings

Q10. What is printed?

```
x = (1, 2, [3, 4])  
x[2].append(5)  
print(x)
```

- A. TypeError
- B. (1, 2, [3, 4])
- C. (1, 2, [3, 4, 5])
- D. (1, 2, 5)

Q11. Which comparison is guaranteed True?

- A. $0.1 + 0.2 == 0.3$
- B. `float('nan') == float('nan')`
- C. `1 == 1.0`
- D. `1 is 1.0`

Q12. What is printed?

```
x = [10, 20, 30, 40, 50]
print(x[4:0:-2])
```

- A. [50, 30]
- B. [40, 20]
- C. [50, 40, 30]
- D. [30, 10]

Q13. What is the type of the result?

```
x = 5 / 2
```

- A. int
- B. float
- C. bool
- D. complex

Q14. What is printed?

```
a = [1, 2, 3]
b = a
b = b + [4]
print(a, b, a is b)
```

- A. [1,2,3,4] [1,2,3,4] True
- B. [1,2,3] [1,2,3,4] False
- C. [1,2,3,4] [1,2,3] False
- D. [1,2,3] [1,2,3] True

Q15. Which statement about tuple hashing is correct?

- A. Every tuple is always hashable.
- B. A tuple is hashable when its elements are hashable.
- C. Tuples cannot be compared.
- D. Tuples are mutable internally.

Q16. What is printed?

```
x = [1, 2, 3]
a, *b = x
print(type(b), b)
```

- A. list [2,3]
- B. tuple (2,3)
- C. list [1,2,3]
- D. tuple (1,2,3)

Q17. What is printed?

```
x = 'abc'
y = x
x += 'd'
print(x, y, x is y)
```

- A. abcd abcd True
- B. abcd abc False
- C. abc abcd False
- D. abc abc True

Q18. Which expression has type `tuple`?

- A. (1)
- B. [1,]
- C. 1,
- D. {1}

Q19. What is printed?

```
x = [[0] * 2] * 3
x[0][0] = 9
print(x)
```

- A. [[9,0],[0,0],[0,0]]
- B. [[9,0],[9,0],[9,0]]
- C. [[0,0],[0,0],[9,0]]
- D. [[9,9],[9,9],[9,9]]

Q20. Which operation preserves the original list object?

- A. $x = x + [5]$
- B. $x = x * 2$
- C. $x.append(5)$
- D. $x = x[:]$

Q21. What is printed?

```
x = (1, 2, 3)
a, *b, c = x
print(a, b, c, type(b))
```

- A. 1 [2] 3 list
- B. 1 (2) 3 tuple
- C. 1 [2,3] None list
- D. 1 [2] 3 tuple

Q22. Which is true about `bool` and `int`?

- A. bool and int are unrelated types.
- B. bool is a subclass of int.
- C. int is a subclass of bool.
- D. True is not numerically equal to 1.

Q23. What is printed?

```
x = [1, 2, 3, 4]
print(x[::2] + x[1::2])
```

- A. [1,2,3,4]

B. [1,3,2,4]

C. [2,4,1,3]

D. [1,3,4,2]

Q24. Which is the best description of `==` versus `is`?

A. Both always test identity.

B. `==` tests value equality; `is` tests object identity.

C. `==` tests identity; `is` tests value equality.

D. They are interchangeable for all data types.

Q25. What is printed?

```
x = (1, [2, 3])
```

```
y = x
```

```
x[1].append(4)
```

```
print(x == y, x is y)
```

A. False False

B. True False

C. True True

D. False True